Reliability Assessment of the Brazilian Power System Using Enumeration and Monte Carlo

Andrea M. Rei and Marcus Theodor Schilling, *Fellow, IEEE*

***Abstract*—Two methods have been largely studied and used in power systems reliability assessment: contingency enumeration and nonsequential Monte Carlo simulation. Both have their advantages and drawbacks. Contingency enumeration is conceptually simple and usually requires low computational effort. Conversely, Monte Carlo simulation is computationally harder, but much more versatile to model random aspects. This paper depicts some important practical aspects regarding the application of both methods, emphasizing how they can be used in a complementary way. The Brazilian interconnected electrical system is used to illustrate the risk assessment of an actual large scale power system, utilizing both techniques.**

***Index Terms*—Monte Carlo method, reliability, state enumeration.**

I. Introduction

TRADITIONALLY, performance of power systems is evaluated using deterministic methods. The most popular method is known as $\mathrm{N}-1$ criterion, in which each outage event in a contingency set results in system performance that satisfies evaluation criteria. Deterministic criteria have achieved reasonable success, but they may lead to quite expensive expansion plans, if fully applied. Furthermore, the contingency set is based on planners and operators experience, and may not cover all critical situations. In practice, real power systems cannot fulfill the $\mathrm{N}-1$ criterion, especially at developing countries, or in case of radial topologies [1]. Then, there is a clear gap between the performance of the planned and the existing systems.

To overcome some deficiencies of deterministic methods, probabilistic criteria may also be used. In doing so, important information is added up to traditional deterministic methods.

Two main techniques are used to evaluate probabilistic reliability assessment of bulk power systems: contingency enumeration for one, two or more transmission and/or generation outages, and Monte Carlo simulations. Both methods have their advantages and drawbacks. Therefore it is important to know how and when to use either method so they can complement each other.

Contingency enumeration for single outages is a simple and usually fast evaluation process and gather much information, specially if it takes outages probabilities into account. However, depending on the power system, it may be necessary to evaluate simultaneous outages of two or more components, either due to the high probability of such events or due to the severe impact over reliability levels. These may be a cumbersome procedure, as simultaneous outages are usually derived from a combinatorial process. In such situations, the performance of Monte Carlo simulations is much better: they are faster and there are no limitations for the number of components in simultaneous outages.

For the last seven years, the Brazilian independent system operator (ISO) has been successfully evaluating the probabilistic reliability level of the interconnected Brazilian transmission system, considering the short-term planning horizon (three-years ahead plan). Based on this experience, this paper will show some actual results of both contingency enumeration and Monte Carlo simulation applied to the Brazilian power system, and the importance of using both techniques in a complementary way. Recognizably, the literature of this subject is exceedingly rich [15], and useful material has been previously published in [23]. However it is hoped that the discussion presented herewith will further confirm results already in the public domain which are, however, scattered in the available literature.

II. Deterministic and Probabilistic Approaches

In order to have a system virtually free of faults, infinite investments would be necessary. This is impossible in a real world, thus it is necessary to establish a compromise between reliability and economy: faults are accepted, since damage is accepted by users and customers. Acceptance levels are quantitatively defined by reliability criteria, which may be deterministic and/or probabilistic.

Deterministic criteria are largely used for transmission expansion planning, and the most popular is the so-called $\mathrm{N}-1$ criterion [1]. The central idea of such procedure if to determine a power system that supports all single emergencies, without operative violations, as partial or total loss of load, overflows and voltage violations. As it is not feasible to evaluate all single emergencies, usually only the most severe ones are considered, based on the experience of planners and operators.

Although conceptually easy and used successfully up to now, deterministic criteria may lead to quite expensive expansion plans. Also, these criteria cannot guarantee the same performance level throughout the system, as they are not able to consider random behavior of components and system operating states, and hence not pondering their effects and consequences.

Unlike deterministic criteria, probabilistic approaches are naturally able to take into account inherent unplanned situations, as load variations and component unavailabilities.

Manuscript received September 27, 2007; revised February 21, 2008. This work was supported in part by the Brazilian governmental agency CNPq and in part by FAPERJ/PRONEX. Paper no. TPWRS-00688-2007.

A. M. Rei is with Cepel, Rio de Janeiro, Brazil (e-mail: andrea@cepel.br).

M. T. Schilling is with the Department of Electrical Engineering, Fluminense Federal University, Rio de Janeiro, Brazil (e-mail: schilling@ic.uff.br).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TPWRS.2008.922532

0885-8950/$25.00 © IEEE
Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on May 08,2025 at 23:02:29 UTC from IEEE Xplore. Restrictions apply.

Besides, consequences and effects of such situations are considered as a proportion of their probability of occurrence. As a consequence, it is possible to determine risk measurements and use them as criteria.

Advantages of probabilistic approaches are noticeable, and they are increasingly being used in transmission expansion planning. Most commercial probabilistic reliability programs currently available address only adequacy aspects, while the assessment of probabilistic security failure modes are still in developmental stages.

III. Probabilistic Reliability Evaluation

Probabilistic reliability assessment of bulk power systems usually comprises the following main steps [15]:

– *selection of system states (contingency states);*

– *evaluation of selected states;*

– *computation of reliability indices and other statistics.*

In the first step, system states are selected and characterized by the operating state of each individual component. For generation systems evaluation, for instance, only generating units are selected. On the other hand, for plain transmission assessment, generation is considered fully reliable and only outages in transmission circuits are considered. Composite assessment comprises outages on both generation units and transmission circuits. Two techniques are most commonly used for state selection, and will be discussed in next section: contingency enumeration and Monte Carlo simulation.

In the second step, each selected state is then evaluated in order to identify adequacy breaches. For transmission and composite adequacy assessment, states are evaluated by power flow solution algorithms, and remedial action procedures may be necessary to eliminate violations, such as overflows and under/over voltages. Many different approaches have been used, ranging from user defined remedial actions to automatic optimal power flow solutions.

In the third step, indices are traditionally computed based on the results of failure states, i.e., states with load curtailment. The most commonly calculated indices, mentioned in the literature [2], [3], are loss of load probability (LOLP), loss of load expectation (LOLE), and expected energy not supplied (EENS).

The three basic steps are repeated until a stopping criterion is satisfied, which depends on the state selection method being used: enumeration or Monte Carlo.

IV. Reviewing State Selection Alternatives

A state of a power system is defined as the combination of the individual states of each component, like generation and transmission equipments and loads, and may be represented by a vector like

$$z = (x_1, x_2, \ldots, x_n) \tag{1}$$

where x_i is the operating state of the ith component. The set of all possible states, arising from combinations of component states is denoted as the *state-space* Z.

Ideally, all possible contingencies states, at all levels, should be evaluated in order to precisely calculate the reliability level of a system. Supposing a system with m components, and two possible operating states for each one, the number of contingency states N_c is equal to

$$N_c = 2^m - 1. \tag{2}$$

From (2) it is clear that it is not practical to select and evaluate all possible contingencies for real bulk systems. For instance, regarding the current Brazilian main grid, the evaluation of all possible double contingencies requires the assessment of 1 821 186 cases and takes approximately 35 days of a dedicated 3.0-GHz computer. This huge number of cases includes the full grid, plus all single and all double contingencies. Each one of all $(\mathrm{n}-1)$ and all $(\mathrm{n}-2)$ cases were evaluated using an interior point ac optimal power flow, considering both voltages, reactive limits and line flows constraints. The interior point algorithm was set with a stringent tolerance, aiming to investigate the utmost potential limits of the algorithm. In this experiment, only one load level was considered [16]–[22]. Two main techniques are then used to select contingency states from the entire state-space: contingency enumeration and Monte Carlo simulation.

A. Contingency Enumeration

In contingency or state enumeration technique, states are selected based on a predefined contingency list, which may comprise the whole system or just a subset. Combining the elements of the list, different contingency levels may be generated, from single to the nth level, in case the list comprises the whole system. However, this only feasible for very small systems. As this procedure may lead to a huge number of states, some cutoff criteria must be used, either isolated or combined, as for instance:

- — *strict contingency level criterion*: in this case the process enumerates and evaluates only all contingencies of predefined levels or depths, like single, double and so on;
- — *probability criterion*: in this case the process evaluates only contingencies with probabilities greater than a predefined threshold;
- — *selected contingencies list criterion*: in this case an *ad-hoc* list is built, comprising, for instance, a set of all single outages plus a limited set of higher order contingencies, such as those representing lines on the same row or lines connected to specific stations.

Enumeration method is straightforward, rather simple to understand and implement on computational programs. In most cases, it is very efficient for single contingencies, or when the state-space is relatively small. Computational effort may increase rapidly if higher contingency levels are considered or when outage probabilities are high. The stopping criterion for contingency enumeration is obvious, when it is based on a predefined contingency list. So state selection process continues until all predefined emergencies have been evaluated.

Reliability indices are obtained combining results from the state evaluation step by the probability of occurrence of each selected state.

Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on May 08,2025 at 23:02:29 UTC from IEEE Xplore. Restrictions apply.

Once it is rather impossible to enumerate all contingency states in real power systems, calculated indices represent lower bounds of real values. This means that all indices will not be less than the calculated values, but may be higher. Thus, the greater the number of contingencies evaluated, the closer results will be to real values. At some extent, contingency enumeration is very similar to deterministic methods like the well-known $N-1$ criterion, when, for instance, contingency depth is limited to first level.

B. Monte Carlo Simulation

Monte Carlo simulation [4], [5] works by sampling a system state based on its joint probability distribution. Usually components may be considered to be statistically independent, and therefore system states are determined by sampling from probability functions of each component. If this is not the case, either statistical data representing dependence should be available, or specific dependence rules should be defined. Since this information is seldom available in practice, it follows that infeasible states may be automatically generated. When this happens, those states must be discarded and the remaining states probabilities must be corrected.

Unlike contingency enumeration, results are no longer lower bounds, but estimates of the real performance of the system. Once the state of the system is a random vector, as in (1), all estimated results, including reliability indices, are also random variables, with associated variances. It is then possible to calculate a relative uncertainty of the estimates, or *coefficient of variation*, given by

$$\beta = \frac{\sqrt{V(\tilde{E}(f))}}{\tilde{E}(f)} \tag{3}$$

where $\tilde{E}(f)$ is an estimated index and $\boldsymbol{V}$ refers to the variance [5].

This parameter is updated during the processes, based on every new estimate of reliability indices. Thus, it is often used as a stopping criterion: the sampling process stops once the coefficient reaches a predefined value, typically 3% or 5%.

Computational effort of Monte Carlo simulation does not depend on system size or complexity. However it is greatly affected by the accuracy required. In order to reduce computational effort, variance reduction techniques may be used, as generalize regression or importance sampling [4]. System performance also affects computational effort, as the number of samples required to estimate a small LOLP, for instance, is greater than what is required for a larger index.

V. Brazilian Power System and Computational Tool

Brazil has 8 547 403 km^2 and more than 170 million inhabitants. The power system interconnects the country from north to south, supplying 98% of the total demand. The remaining 2% are supplied by isolated systems.

Regarding the main ac transmission network, voltages range from 230 kV to 750 kV, totalizing 80 022 km of transmission lines and transformers. As shown in Fig. 1, a 600-kV dc link is also present. Peak demand for year 2004 was equal to 56 795 MW. The southern part of the country is responsible for near 80% of total generation and load, and the northern part for the other 20%.

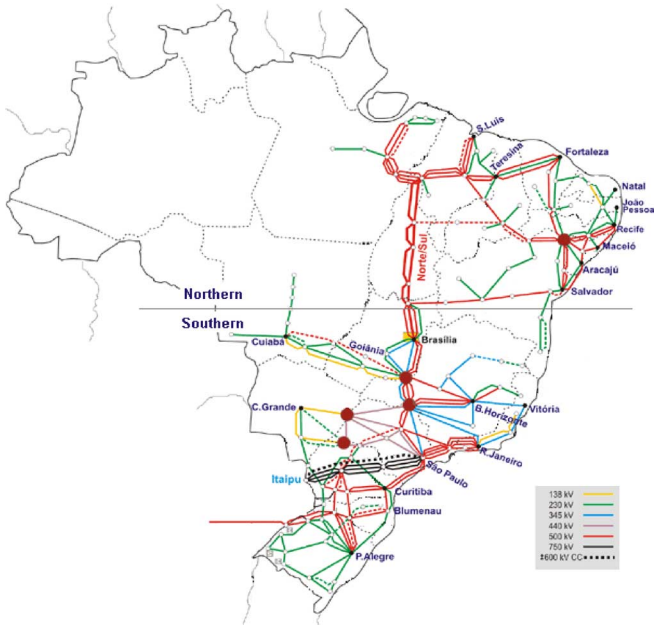


Fig. 1. Brazilian main transmission network.

The Brazilian system used in this paper is based on a future configuration (year 2008). To model the system, there are approximately 3800 buses and 5500 circuits, including 2000 transformers [7]. Generation system has approximately 680 units, and maximum capacity of 64 600 MW. The future system has a total estimated load equal to 61 800 MW. Electrical model of the system comprises voltages from 13.8 kV to 750 kV, and individual generating unit capacity ranges from 1 MW to 720 MW. Random behavior of transmission circuits and generation units is represent by two-state Markovian models [3]. Outage data—failure and repair rates—are based on the past performance of the system [8], [9]. Outage data from international sources available in [10] and [11] were also considered in some comparative analyses. Transmission outage data are considered only for the main network, i.e., above 230 kV. Generation system state-space is composed of generating units greater than 40 MW. All remaining elements were considered fully reliable (that means that no outage probability was ascribed to them). This option was chosen in order to improve the optimal power flow computational efficiency (when all generating unities are considered, including those unities below 40 MW, the computational burden is much higher, without the benefits of better results.

The computational program [12], [13] used in this study was designed for bulk power system composite reliability assessment, but also comprises other deterministic and probabilistic evaluations. Its main features are:

- ac network modeling;
- power system solution environment comprising:
  - — power flow solution, by full Newton–Raphson;
  - — ac optimal power flow, based on interior point method, with minimum load shed objective function;

Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on May 08,2025 at 23:02:29 UTC from IEEE Xplore. Restrictions apply.

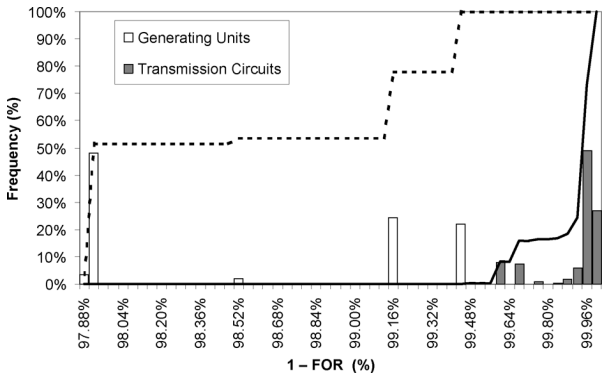


Fig. 2. Statistical profile of Brazilian transmission circuits and generating units.

— individual contingency analyses;
— contingency list analyses;
— probabilistic power flow;

- optimal power flow used for corrective measures;
- random behavior of generating units and transmission circuits are represented by Markovian models—two and multiple state models are available;
- common mode failures;
- reliability assessment by contingency enumeration or Monte Carlo simulation;
- traditional and frequency and duration reliability indices, stratified in three levels: system, area, and buses;
- additional statistics: probability density functions of selected variables (power flows, voltages, etc.), failure mode indices, etc.

VI. Random Behavior of Generating Units and Transmission Circuits

In most systems, random behavior of generating system is very different from the behavior of transmission system, as transmission circuits are usually more reliable than generating units.

This important difference is depicted in Fig. 2 for the studied Brazilian system and its original outage data. Almost all transmission circuits' availabilities are greater than 99.6%, whereas the availability of generating units is not greater than 99.5%. In this example, representing outage data for only 370 generating units and 1750 circuits, the probability that all circuits are simultaneously available is approximately 14.42%. (i.e., considering the grid without contingencies). On the other hand, the probability that all generating units are simultaneously available is around 0.1051%, or almost 140 times less than the probability associated to the transmission network. Therefore, in the particular case of the Brazilian system, it is verified that the probability of the full transmission system state (i.e., that state with all transmission lines in operation) would only equal the probability of the full generation system state if there were approximately 6000 circuits.

A similar behavior is observed applying outage data available from international sources [11] to this system. For instance, considering transmission data from [10], the probability that all circuits are simultaneously available is 2.0%. For the generation system, considering outage data of one of the best performance generating units in [11] (i.e., lowest forced outage rate), and applying it to all units in the Brazilian system, the probability that all generating units are simultaneously available is just 0.0026%. In this case, the difference in the behavior of transmission and generation systems is even greater (almost a 770 times ratio).

TABLE I
Probabilities (%) Associated to the Brazilian Transmission and Generation Systems

	Brazilian Statistics	
	All Elements Up	All Single Contingencies
Generation	0.1051	0.7407
Transmission	14.42	27.92

The comparison of generation and transmission system availabilities is presented in Table I. Considering original outage data of the Brazilian system, it can be seen that a traditional contingency enumeration method of single transmission emergencies would cover only approximately $(14.42 + 27.92 = 42.34\%)$ of the total state-space. Double and higher order transmission contingencies represent a great amount of the state-space $(100 - 42.34 = 57.66\%)$, which in some cases might not be neglected. In the actual Brazilian system, second order transmission contingencies represent around 27.05% of the state-space, and in order to consider their effects (and then 69.39% of the state-space) it would be necessary to evaluate over 1.8 million system states.

The performance of the generation system is quite worse, since using the enumeration technique it would not be possible to cover more then 4% of the state-space, even considering second order contingencies. Under such situations, Monte Carlo simulations might be the alternative solution. Therefore, depending on the uncertainty of the estimates, Monte Carlo could successfully cover a trustworthy amount of the total state-space, with less computational effort.

VII. Reliability Assessment Results

Reliability of the actual Brazilian interconnected power system was assessed using both contingency enumeration and Monte Carlo simulation. For all tests, the same network configuration, generation capacity and load were used.

Transmission and generation systems were evaluated separately and also combined. Thus, although the test system is exactly the same, the state-space varies, modifying the probability of the base case and contingency states.

For transmission assessment, the probabilistic state-space is composed of 1750 elements (transmission lines and transformer). The probabilistic state-space for generation system assessment is composed of 370 generating units which are geographically located at 140 different power stations scattered in the country. Some power stations have just one generation unit, while in the other extreme, one particular station has up to

Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on May 08,2025 at 23:02:29 UTC from IEEE Xplore. Restrictions apply.

TABLE II
STATE–SPACE COVERAGE FOR SINGLE CONTINGENCY ENUMERATION—BRAZILIAN SYSTEM

System	*Probabilities (%)*			*Computing Time (pu)*
	Base Case (1)	*All Single Contingencies (2)*	*State Space Evaluated (1+2)*	
Transmission	14.42	27.92	42.34	1.0 (*)
Generation	0.1051	0.7407	0.8458	0.1
Generation and Transmission	1.45	6.25	7.70	1.1

(*) Computing time for single transmission contingency enumeration is used as the base to calculate computational effort for the other tests.

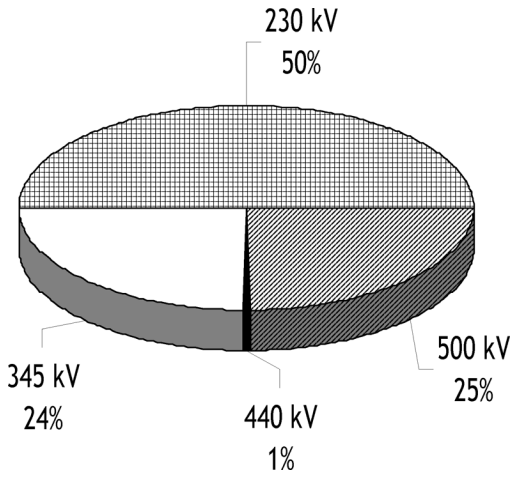


Fig. 3. Contribution of each voltage level in reliability results—Brazilian system.

20 units. Composite assessment is carried out considering the same elements used in the independent evaluations.

A. Contingency Enumeration

Contingency enumeration evaluation considered only single outages, and the basic results showing the state-space coverage are shown in Table II. Probability threshold was not used as a cutoff criterion. Reliability indices and computer times of simulations are given in per unit or percent values in order to emphasize the relativity of measures and the hardware independent aspect.

If outage data are considered only for transmission circuits and generating system is taken as fully reliable, contingency enumeration method can cover little less than 50% of the state-space (actual value is 42.34%). Single transmission outages lead to adequacy breaches in 19% of the evaluated emergencies. Load curtailment is needed in 13% of the evaluated emergencies.

Single contingency enumeration considering outage data only for generating units equal to and above 40 MW, is much faster, due to the small number of emergencies being evaluated—only 127 generating units meet this criterion. Single generation outages lead to adequacy breaches in only 10% of the emergencies. However, they are effectively eliminated by remedial actions, such as generation redispatching. Therefore, there is no load curtailment for single generation outages. As stated before, generating units have higher unavailability, compared to transmission circuits. Thus, only a small portion (less than 1%) of the state-space could be covered (actual value equal to 0.8458%). This result indicates that, in the Brazilian case, single contingency enumeration of the generation system is not enough to capture relevant failure modes. This result also hints at the relevance of an accurate generation maintenance representation.

As was discussed in the previous section, availabilities associated to generation systems are much smaller than those of transmissions systems. As a result, in the generation and transmission analysis, the generation system "dominates" the state-space, greatly reducing probabilities associated to base case and single contingencies. Table II also shows that only 7.7% of the composite state-space could be evaluated. It is also seen that the computing effort is very similar to the transmission assessment (1.1 p.u.).

Although covering just 42.34% of the transmission state-space, probabilistic single contingency enumeration yet gives a variety of information that would not be available with a non-exhaustive deterministic evaluation. Some important questions may be easily answered by a probabilistic contingency enumeration, such as the following.

- *Does the system actually meet the $N-1$ criterion?*
- *Are there $N-1$ regions/areas? Which?*
- *How are reliability levels distributed throughout the system, regarding its areas, regions, and buses?*
- *How do different failure modes, as overloads, islanding voltage violations, and generation deficits, affect reliability levels?*

For transmission enumeration, load curtailment is needed in 13% of all single contingency states, so it can be said that the studied system is not $\mathrm{N}-1$.

In Fig. 3, in can be seen that the greatest contribution (50%) to reliability performance comes from the 230-kV network, despite it represents only 33% of the state-space (see Fig. 4). The 500-kV network has almost the same size regarding the state-space (34%, see Fig. 4), however, as shown in Fig. 3, its contribution to reliability indices (25%) is half of that of the 230-kV system. This results reflects the higher reliability of the 500-kV grid.

On the other hand, 750- and 440-kV networks cover a small part of the state-space (4% and 3%, respectively, see Fig. 4) but may be practically considered $\mathrm{N}-1$ adherent, since their contribution in Fig. 3 is nil.

Thus, comparing Figs. 3 and 4, it can be clearly seen that the contribution of each subsystem to reliability indices does not follow their contribution to the state-space.

Another important information concerns the contribution of different failure mode, as overloads, voltage violations, islanding, to adequacy breaches of reliability indices, as shown in Fig. 5.

It may be argued that much of these information may also be obtained from traditional deterministic approaches, depending

Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on May 08,2025 at 23:02:29 UTC from IEEE Xplore. Restrictions apply.

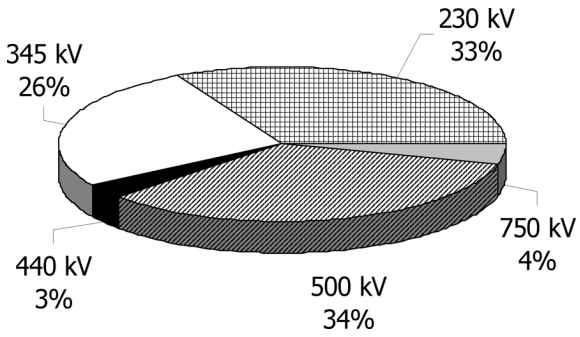


Fig. 4. Voltage level contribution in the evaluated state-space—Brazilian system.

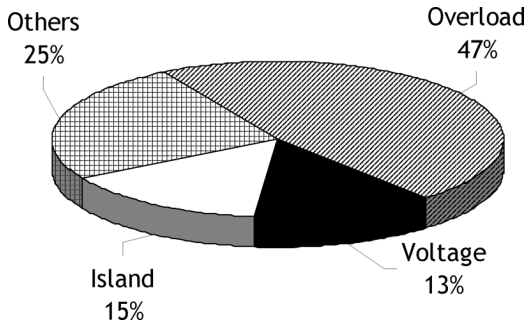


Fig. 5. Contribution of failure modes in adequacy breaches—Brazilian system.

TABLE III
RESULTS FOR MONTE CARLO SIMULATION—BRAZILIAN SYSTEM

System	*Probabilities (%)*				*Computing Time (pu)* (*)
		Contingencies			
	Base Case	All Single	All Double	Higher Order	
Transmission	14.42	27.92	27.05	30.61	3.37
Generation and Transmission	0.0	0.24	0.60	99.15	8.47

(*) Computing time for single transmission contingency enumeration is used as the base to calculate computational effort - see Table I

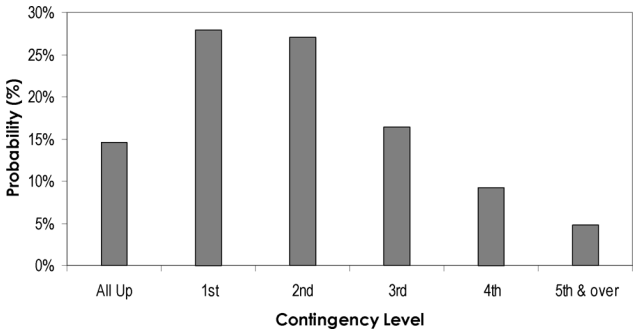


Fig. 6. Contingency level distribution in transmission assessment by Monte Carlo simulation—Brazilian system.

on the experience of the planner or operator. However, single contingency enumeration enables the evaluation and analysis of all $N-1$ topologies, providing information in a more organized and detailed way [1]. These information may be used to better redirect economic resources and reduce the gap between weaker and stronger areas.

B. Monte Carlo Simulation

As Monte Carlo simulation [4]–[6] works on sampling from the entire state-space, not only evaluating some predefined contingencies, it can easily cover situations and configurations not considered in contingency enumeration. Transmission and composite reliability evaluation were then carried out using this state selection method.

Results presented in Table III were obtained for a coefficient of variation equal to 5% for the EENS index. For transmission reliability, computing time is approximately 3.37 times the effort associated to single contingency enumeration. However, it is much less than the effort necessary to evaluate all double contingencies with a state enumeration method (approximately 35 days for the current Brazilian system). It is also noticed that the Monte Carlo composite evaluation, although more time consuming than the single transmission contingency enumeration, by a factor of 8.47, is not only feasible in practice, as more numerically precise, since a 5% coefficient of variation is usually attainable.

Contingency level distribution estimative can also be calculated using Monte Carlo simulation technique, as presented in Fig. 6. For transmission assessment, it can be seen that the probability of occurrence of the base-case and first order contingencies is very similar to the results obtained for contingency enumeration. Although second and higher order contingencies represent around 57.66% of the state-space, the decision in order to consider them or not depends on the system and their impact on reliability indices.

TABLE IV
TRANSMISSION ASSESSMENT INDICES—BRAZILIAN SYSTEM

	LOLP	*EENS*	*Severity Index*
Enumeration (*)	1.0	1.0	1.0
Monte Carlo	6.11	7212.39	7.21

(*) Transmission single contingency enumeration indices are considered reference indices

C. Single Enumeration versus Monte Carlo Simulation

Table IV shows the comparison of reliability indices for transmission assessment using single contingency enumeration and Monte Carlo simulation techniques. This comparison is interesting because it shows the effect of higher outages (double, triple, and higher, obtained through Monte Carlo simulation) on the reliability indices obtained with single outages enumeration. Considering enumeration indices as a reference, it can be seen that the difference of both indices (LOLP and EENS) is high and not uniform. The LOLP index is 6 times higher, while the EENS index is approximately 7000 times higher. The severity index [14], which is the most common index used in Brazil, is seven times higher. In this case, it is obvious that single contingency enumeration results are far from the real risk performance, and must be considered with care. They must be faced as they truly are: a measure of the system performance conditioned to single

Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on May 08,2025 at 23:02:29 UTC from IEEE Xplore. Restrictions apply.

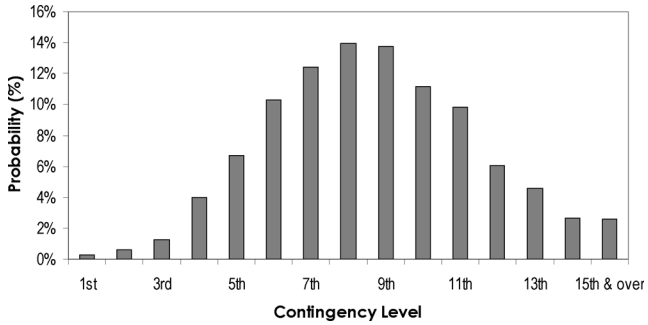


Fig. 7. Contingency level distribution in composite assessment by Monte Carlo simulation—Brazilian system.

TABLE V
COMPOSITE ASSESSMENT INDICES—BRAZILIAN SYSTEM

	LOLP	*EENS*	*Severity Index*
Enumeration	1.0	1.0	1.0
Monte Carlo	5848	6197	6221

emergencies. The difference between single contingency enumeration and Monte Carlo simulation results represents the impact of second and higher order contingency on system reliability, which, in this case, might not be neglected, depending on the analyst purposes.

A similar comparison was carried out for composite assessment. When considering generation outages in the Brazilian system, it is likely to have higher order contingencies, especially from 6th to 11th order, as shown in Fig. 7, which represent around 71% of the state-space.

Due to this characteristic, indices calculated by single contingency enumeration and Monte Carlo simulation are quite different, as shown in Table V. Thus, for composite assessment, state enumeration does not seem to be adequate (even if 3rd- or 4th-order emergencies are considered, which would require a great computational effort) and Monte Carlo simulation might be used instead.

One question regarding Monte Carlo simulation refers to the adequate level of accuracy to be considered. This parameter depends on the system under evaluation and on the needs of planner and operators, once there must be a compromise between accuracy and computational effort. Convergence process is not equal for all indices, due to the differences in their variances; refer to (3).

The LOLP index, for instance, usually has a smaller variance compared to EENS index, so it converges first. All results presented up to now were estimated using a coefficient of variation equal to 5% for the EENS index. In this case, LOLP index was estimated with an uncertainty of 3.44%.

The convergence process of EENS and LOLP indices for the Brazilian system is exemplified in Fig. 8. Coefficient of variation drops quickly in the beginning, reaching 10% with less than 1000 samples and one fourth of the effort needed for a 5% coefficient. Then the process becomes slower, and almost 10 000 samples are necessary to reach a 3% coefficient.

The LOLP index seems to stabilize first, due to its lower variance, while EENS is still oscillating. Better estimates may be

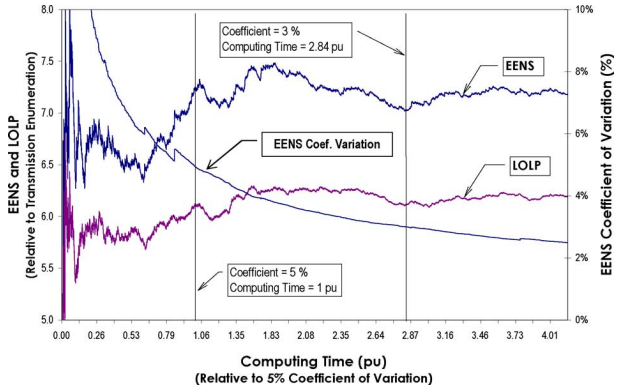


Fig. 8. Convergence of EENS and LOLP indices for the main grid Brazilian power system.

obtained for coefficients lower than 3%, but computational effort is at least three times harder. Estimated indices for a 5% coefficient are very near the results for uncertainties lower than 3%. However, the estimates are quite varying between these two limits, especially the EENS index. For this example, coefficients greater than 5% may lead to doubtable results.

Variance reduction techniques, as suggested in [4], were not used in these tests, but may be used and investigated in order to improve simulation performance.

VIII. CONCLUSION

Nonsequential Monte Carlo simulation and contingency enumeration are the most common state selection techniques used in probabilistic reliability assessment of bulk power systems. Both have their advantages and drawbacks.

State enumeration is easier and the computational effort is usually feasible for single order contingencies. However, evaluations of high order contingencies may be intractable.

On the other hand, Monte Carlo simulation usually requires larger computational effort, but is very versatile to model random behavior of components. It may be the easiest way, if not the only one, to evaluate adequately higher order contingencies. Therefore it is clear that both techniques should be used in a complementary way.

For the Brazilian power system reliability assessment, single transmission contingency enumeration is currently used as the first step of a regular probabilistic analysis. It demands from planners and operators almost the same effort used for the traditional pure deterministic $N-1$ criterion [1]: it is conceptually simple and its computational effort is low. Furthermore, it renders a huge amount of information about the system risks. Contingency enumeration is best suited for systems with low failure probabilities, like the usual transmission systems. In this case, single contingency evaluations may not cover a great amount of the state-space, yet it is possible to determine how far the system is from the $N-1$ criterion, the most prevailing failure modes, and lower bounds of reliability indices.

For generation and composite assessment, where higher failure probabilities are considered, state enumeration may lead to very poor results. In such systems, high order contingencies are highly probable, and difficult to be captured by the enumeration method. Depending on generation capacity available,

Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on May 08,2025 at 23:02:29 UTC from IEEE Xplore. Restrictions apply.

reliability level may be very high, regarding generation failures, and so very hard to estimate. As was shown in this paper, this happens in the Brazilian system. Monte Carlo simulation is most efficient to cover state-spaces composed of unbalanced failure probabilities, like generation units and transmission circuits. Then, it might be the best solution for performing a composite analysis.

When using Monte Carlo simulation, it is very important to choose an appropriate coefficient of variation, as high values may lead to poor results. On the other hand, very low coefficients may result in unnecessary computational effort. For the Brazilian power system, a coefficient of variation of 3% has been feasible to reach within a reasonable computer time. Computationally, Monte Carlo simulation is often harder than state enumeration, but gives information not available in the latter. Furthermore, variance reduction technique may be used to improve its performance. Best results may be achieved combining contingency enumeration and Monte Carlo simulation, in different evaluations that complement each other. In Brazil, this strategy has been used with success, helping to render probabilistic reliability analysis as a useful tool for short-term planning purposes.

REFERENCES

[1] M. T. Schilling, A. M. Rei, M. B. Do Coutto Fo, and J. C. S. Souza, "On the implicit probabilistic risk embedded in the deterministic 'n − α' type criteria," in *Proc. 2002 Int. Conf. Probabilistic Methods Applied to Electric Power Systems*, Naples, Italy, 2002.

[2] PROSD Working Group, IEEE PES APM Subcommittee, "Reliability indices for use in bulk power supply adequacy evaluation," *IEEE Trans. Power App Syst.*, vol. PAS-97, no. 4, pp. 1097–1103, Jul./Aug. 1978.

[3] R. Billinton and R. N. Allan, *Reliability Evaluation of Power Systems*, 2nd ed. New York: Plenum, 1994.

[4] Composite System Reliability Evaluation Methods EPRI Tech. Rep. EL-5178, 1987.

[5] R. Billinton and W. Li, *Reliability Assessment of Electric Power Systems Using Monte Carlo Methods*. New York: Plenum, 1994.

[6] R. Billinton and X. Tang, "Selected considerations in utilizing Monte Carlo simulation in quantitative reliability evaluation of composite power systems," *Elect. Power Syst. Res.*, vol. 69, pp. 205–211, 2004.

[7] ONS, Annual Report—Year 2004 (in Portuguese) Brazil, 2005 [Online]. Available: http://www.ons.org.br

[8] CIER, Sistema de Estatistica CIER – Manual de Geração (in Portuguese) 1990.

[9] ONS, Probabilistic Performance of Transmission Lines in Brazil (in Portuguese) 2001, Tech. Rep. ONS-2.1-033/2001.

[10] Forced Outage Performance of Transmission Equipment. Montreal, QC, Canadian Electricity Association, 1999.

[11] NERC, Generating Availability Report: 2000-2004 NJ, 2005. [Online]. Available: http://www.nerc.com.

[12] J. C. O. Mello, A. C. G. Melo, S. P. Romero, G. C. Oliveira, S. H. F. Cunha, M. Morozowski, M. V. F. Pereira, and R. N. Fontoura, "Development of a composite system reliability program for large hydrothermal power systems – Issues and solutions," in *Proc. 3rd Int. Conf. Probabilistic Methods Applied to Electric Power Systems*, 1991, pp. 64–69.

[13] CEPEL, NH2 Program – User's Guide (in Portuguese)7.3 ed. 2005.

[14] C. C. Fong *et al.*, "Bulk system reliability measurement and indices," *IEEE Trans. Power Syst.*, vol. 4, no. 3, pp. 829–835, Aug. 1989.

[15] R. Billinton, M. Fotuhi-Firuzabad, and L. Bertling, "Bibliography on the application of probability methods in power systems reliability evaluation 1996–1999," *IEEE Trans. Power Syst.*, vol. 16, no. 4, pp. 595–602, Nov. 2001.

[16] S. H. F. Cunha, F. B. M. Gomes, G. C. Oliveira, and M. V. F. Pereira, "Reliability evaluation in hydrothermal generating systems," *IEEE Trans. Power App. Syst.*, vol. PAS-101, no. 12, pp. 4665–4673, Dec. 1982.

[17] C. Singh and Q. Chen, "Generation system reliability evaluation using a cluster based load model," *IEEE Trans. Power Syst.*, vol. 4, no. 1, pp. 102–107, Feb. 1989.

[18] A. M. L. da Silva, J. Coelho, and A. C. G. Mello, "Uncertainty considerations in frequency an duration analysis for large scale hydrothermal generating systems," *Proc. Inst. Elect. Eng.* , vol. 139, no. 3, pt. C, pp. 277–285, May 1992.

[19] S. Granville, "Optimal reactive dispatch through interior point methods," *IEEE Trans. Power Syst.*, vol. 9, no. 1, pp. 136–146, Feb. 1994.

[20] S. Granville, J. C. O. Mello, and A. C. G. Mello, "Application of interior point methods to power flow unsolvability," *IEEE Trans. Power Syst.*, vol. 11, no. 2, pp. 1096–1103, May 1996.

[21] A. C. G. Melo, J. C. O. Mello, and S. Granville, "The effects of voltage collapse problems in the reliability evaluation of composite systems," *IEEE Trans. Power Syst.*, vol. 12, no. 1, pp. 480–488, Feb. 1997.

[22] C. R. R. Dornellas, M. T. Schilling, A. C. G. Melo, J. C. S. Souza, and M. B. Do Coutto Filho, "Combining local and optimized power flow remedial measures in bulk reliability assessment," *Proc. Inst. Elect. Eng., Gen., Transm., Distrib.*, vol. 150, no. 5, pp. 629–634, Sep. 2003.

[23] R. N. Allan, M. P. Bhavaraju, R. Billinton, C. C. Fong, M. V. F. Pereira, R. J. Ringlee, L. Salvaderi, and C. Singh, "Reliability assessment of composite generation and transmission systems," in *IEEE Tutorial Course*, 1990, Course Text 90EH0311-1-PWR.

Andrea M. Rei was born in Rio de Janeiro, Brazil, in August 1966. She received the B.Sc. degree from the State University of Rio de Janeiro (UERJ) in 1988 and the M.Sc. and D.Sc. degrees, all in electrical engineering, from the Catholic University of Rio de Janeiro (PUC/RJ) in 1992 and 1997, respectively.

From 1999 to 2002, she worked at the Brazilian ISO, in charge of the Reliability Studies Task Force. Since 2002, she has been working as a Researcher at CEPEL, coordinating research and deployment of reliability of bulk power systems and the NH2 program.

Marcus Theodor Schilling (M'78–SM'86–F'05) received the B.Sc. degree from the Catholic University of Rio de Janeiro (PUC/RJ), Rio de Janeiro, Brazil, in 1974 and the M.Sc. and D.Sc. degrees, all in electrical engineering from the Federal University of Rio de Janeiro (COPPE/ UFRJ) in 1979 and 1985, respectively. He has worked at Furnas, the Catholic University of Rio de Janeiro, Eletrobrás, Cepel, and ONS (in Brazil), Universität Dortmund, (in Germany), and Ontario Hydro (in Canada). Currently, he is a Professor at the Fluminense Federal University (UFF/TEE/IC), Rio de Janeiro. His main research interests are power systems probabilistic methods and computer applications in power systems.

Prof. Schilling has been Chairman of the Brazilian Reliability Subgroup (SGC) and Manager of the Electrical Studies Division at Eletrobrás. He is a Registered Professional Engineer in Brazil (CREA) and Canada (PEO).

Authorized licensed use limited to: b-on: Universidade de Coimbra. Downloaded on May 08,2025 at 23:02:29 UTC from IEEE Xplore. Restrictions apply.